

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

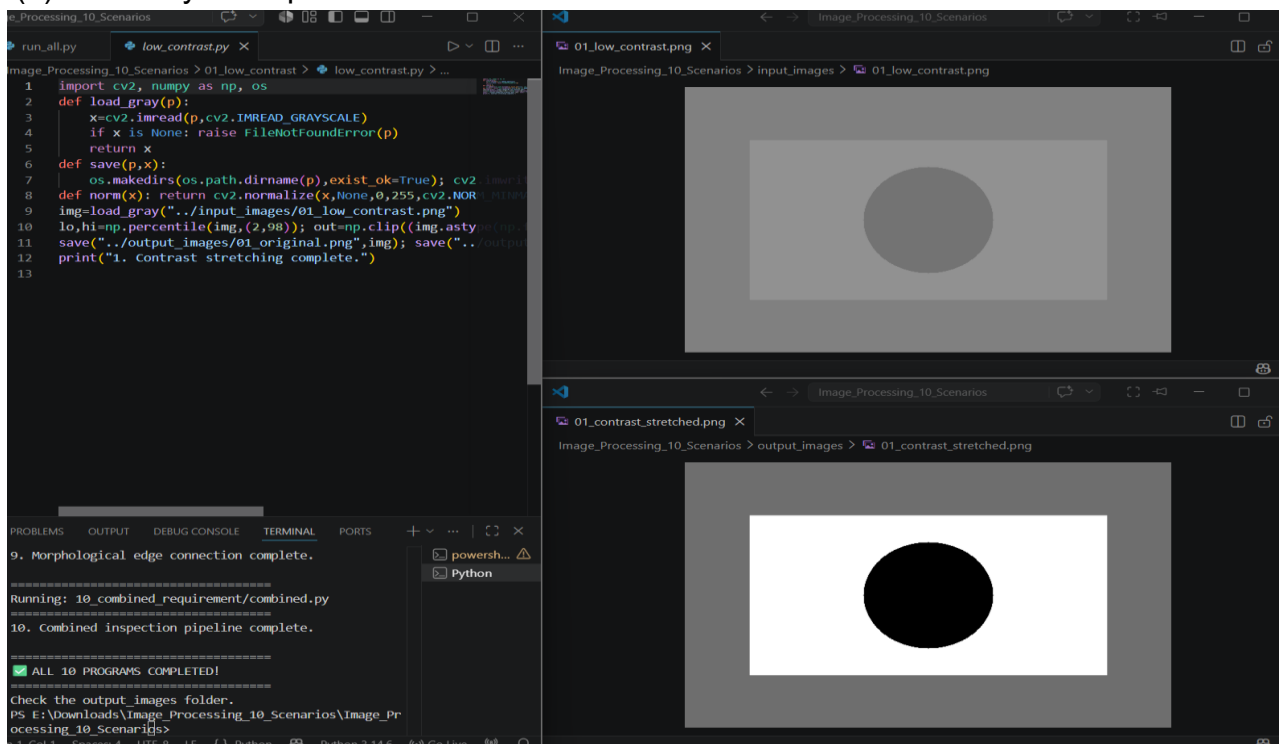
Assessment Tool Weightage: 20%

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

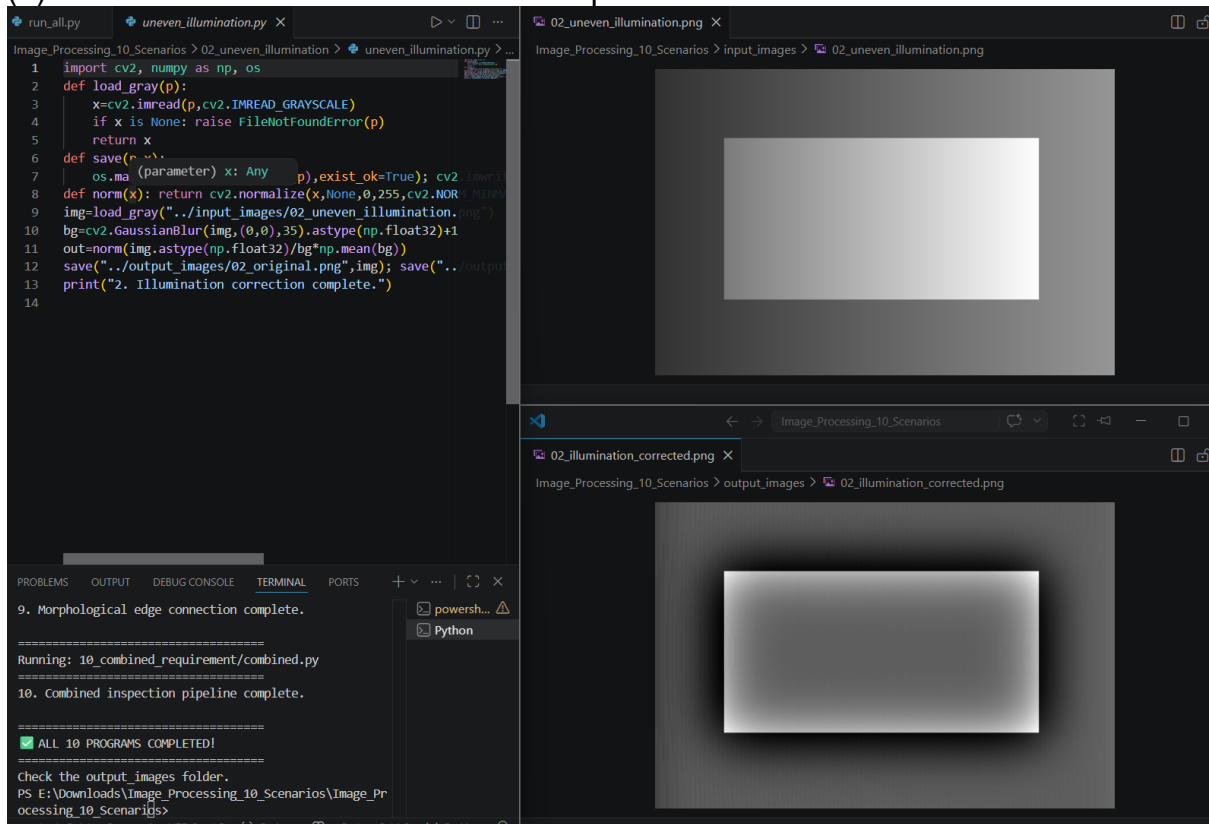
- Clearly interpret the scenario and explain the cause of low contrast.
- Identify all key issues affecting image visibility.
- Apply an appropriate enhancement technique to address the problem.
- Justify your choice with logical reasoning.
- Present your explanation in a clear and structured manner.



2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

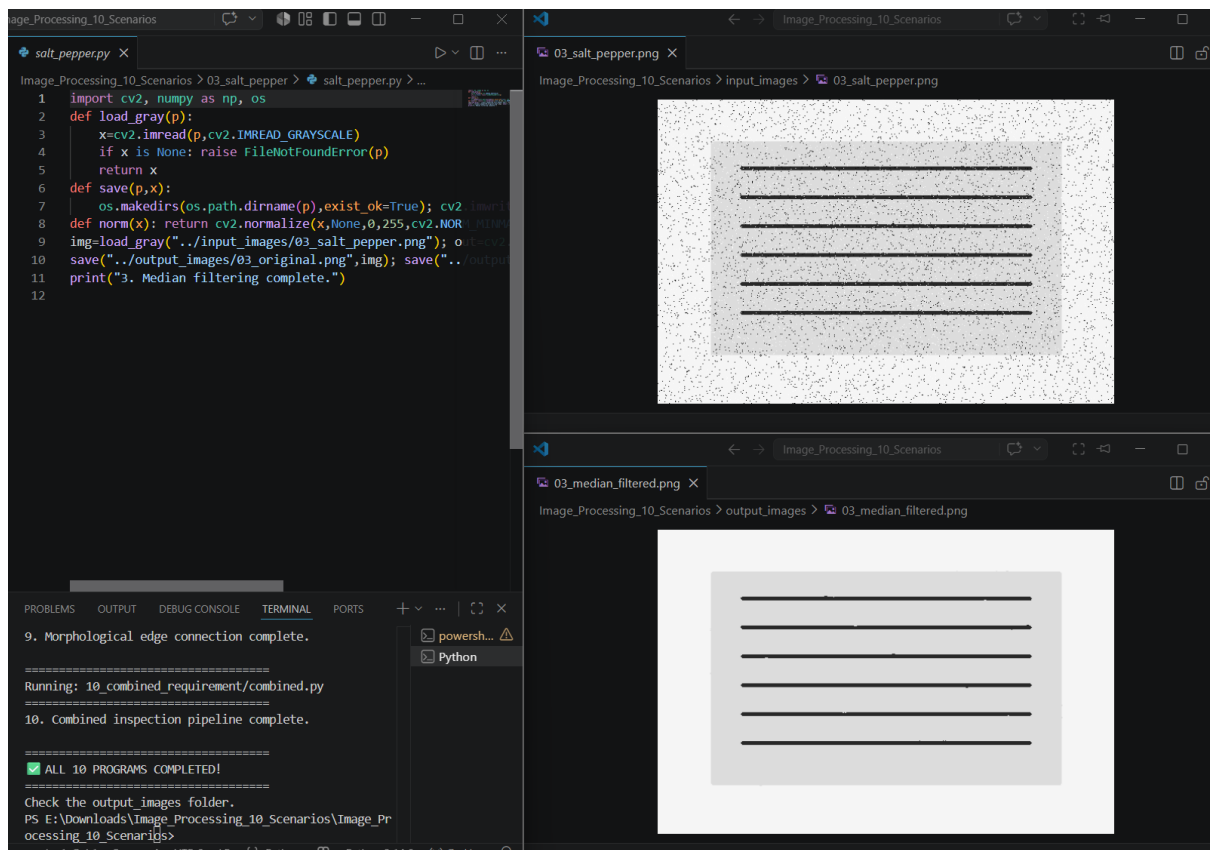
- Interpret the scenario and explain the impact of uneven illumination.
- Identify key factors causing intensity variation.
- Apply a suitable enhancement method to normalize illumination.
- Justify your selected approach with reasoning.
- Provide a clear and well-structured explanation.



3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

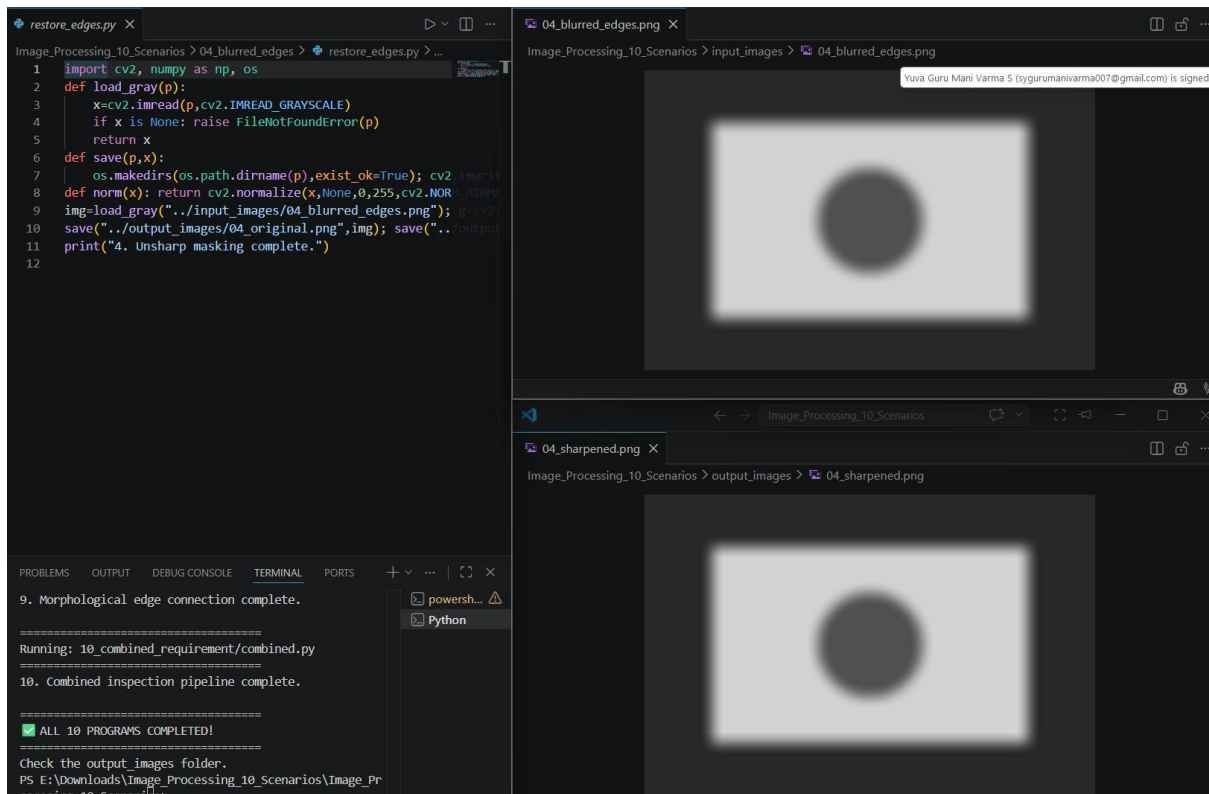
- Explain the scenario and characteristics of this noise.
- Identify key issues impacting image quality.
- Apply an appropriate filtering technique.
- Justify why this method is more suitable than alternatives.
- Present your response clearly and logically.



4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

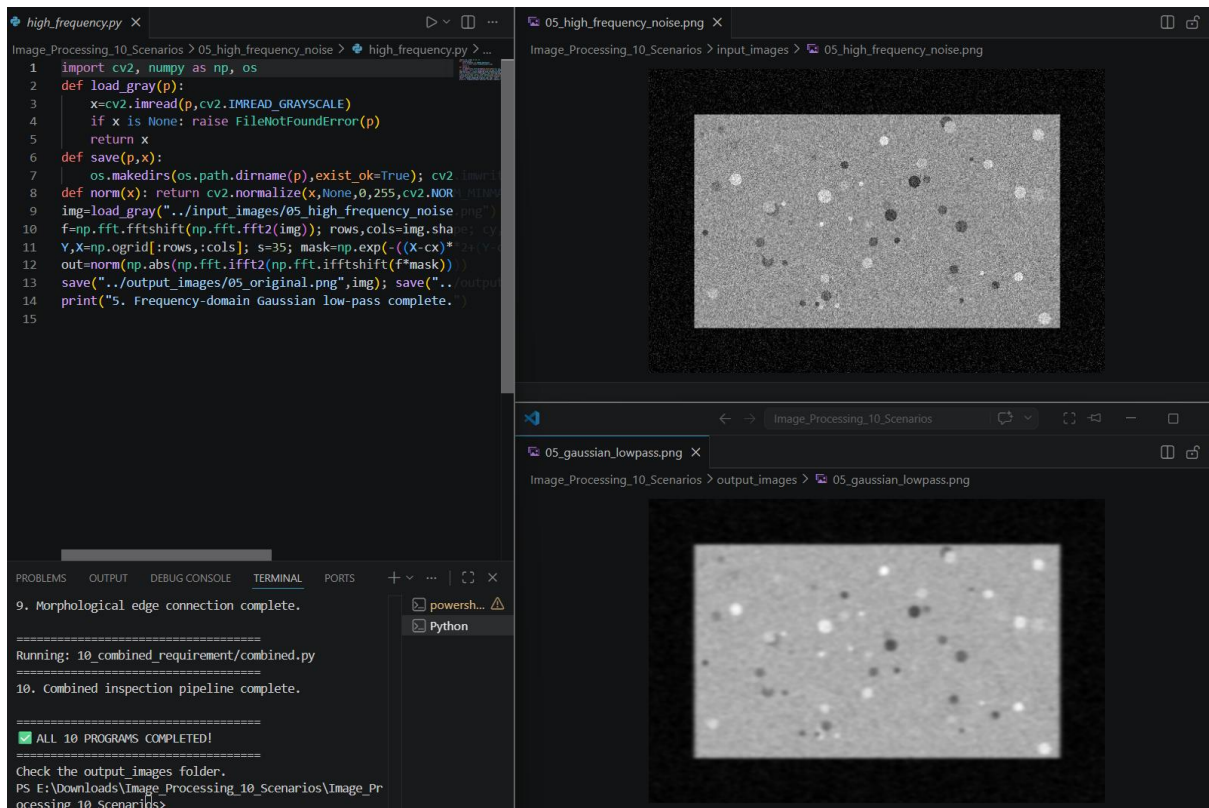
- Interpret the scenario and explain why edges are blurred.
- Identify key issues related to smoothing and edge loss.
- Apply an appropriate technique to restore edges.
- Justify your decision with clear reasoning.
- Structure your explanation clearly.



5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

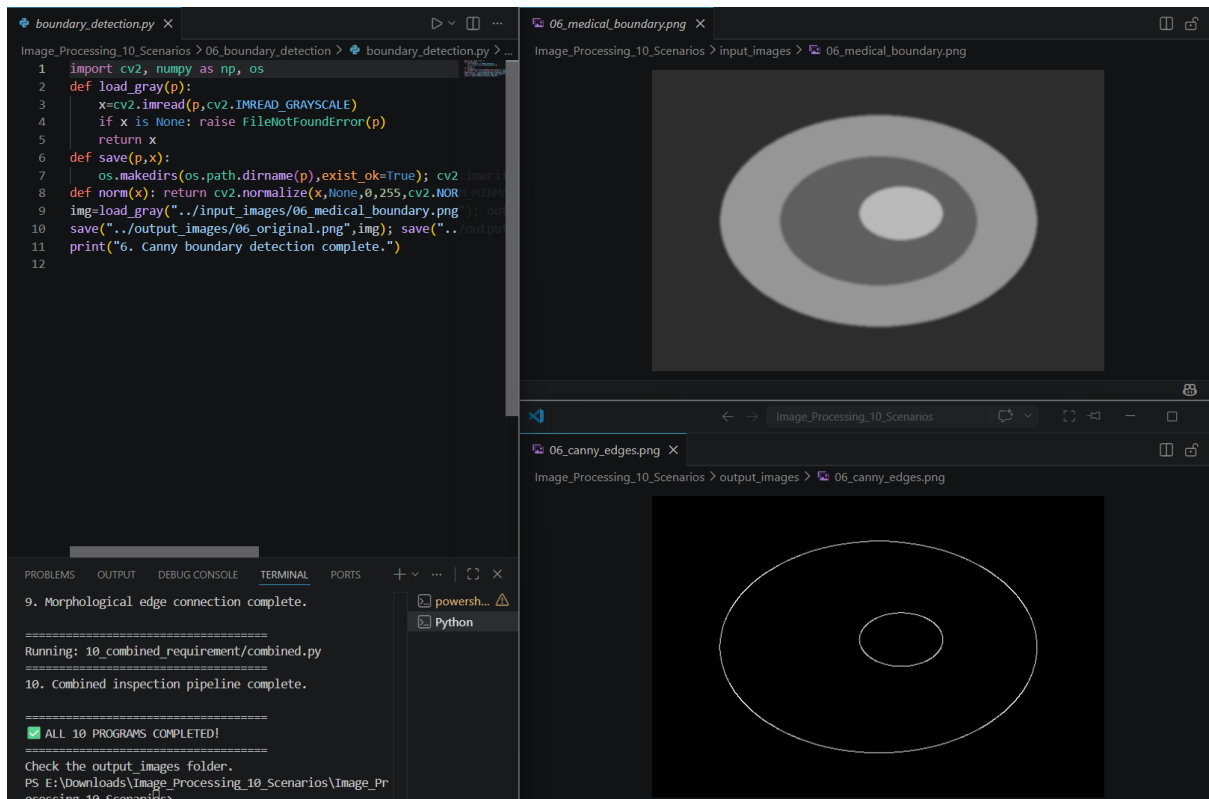
- Interpret the scenario and describe high-frequency noise.
- Identify the key issues affecting image clarity.
- Apply an appropriate frequency domain filtering method.
- Justify the choice of filter with reasoning.
- Present your answer in a clear and organized manner.



6. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

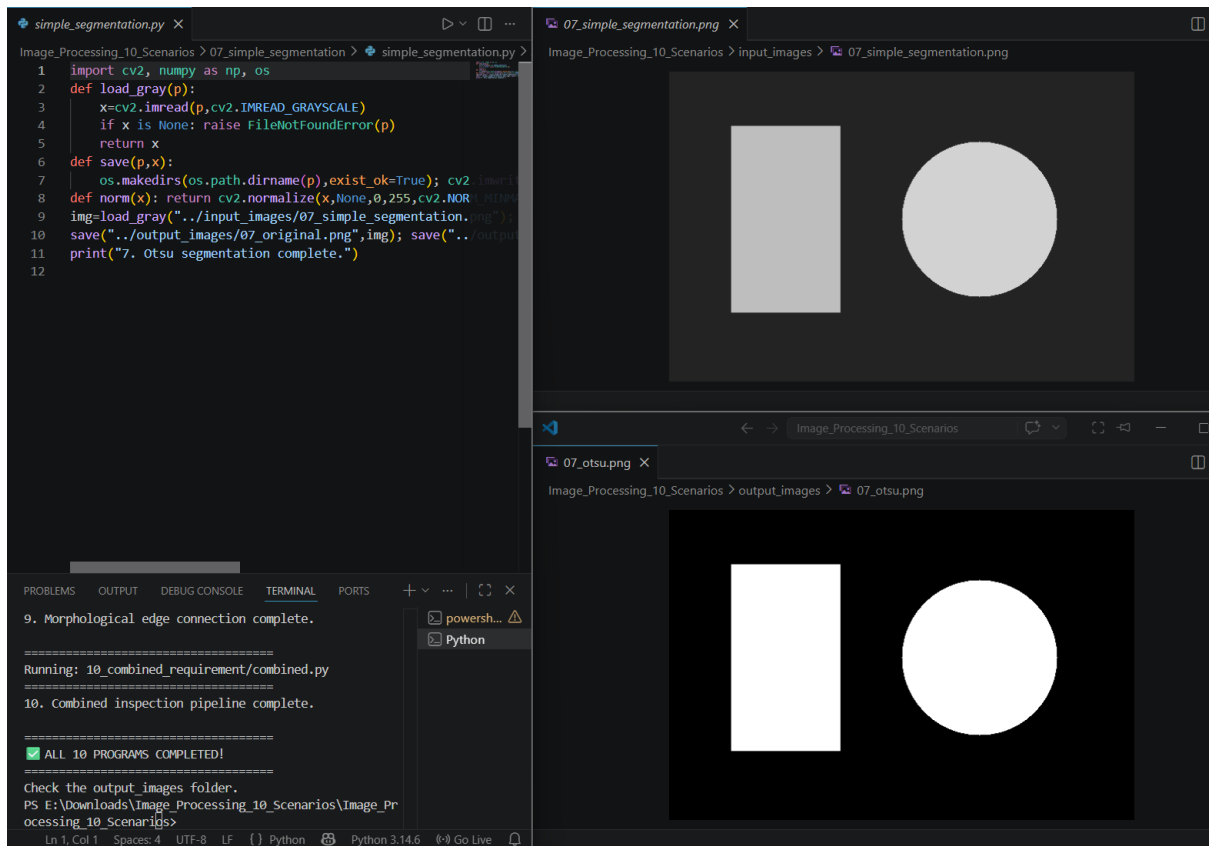
- Interpret the scenario and explain the importance of boundary detection.
- Identify key challenges in detecting boundaries.
- Apply a suitable edge detection method.
- Justify your selection with logical reasoning.
- Present your explanation clearly and systematically.



7. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

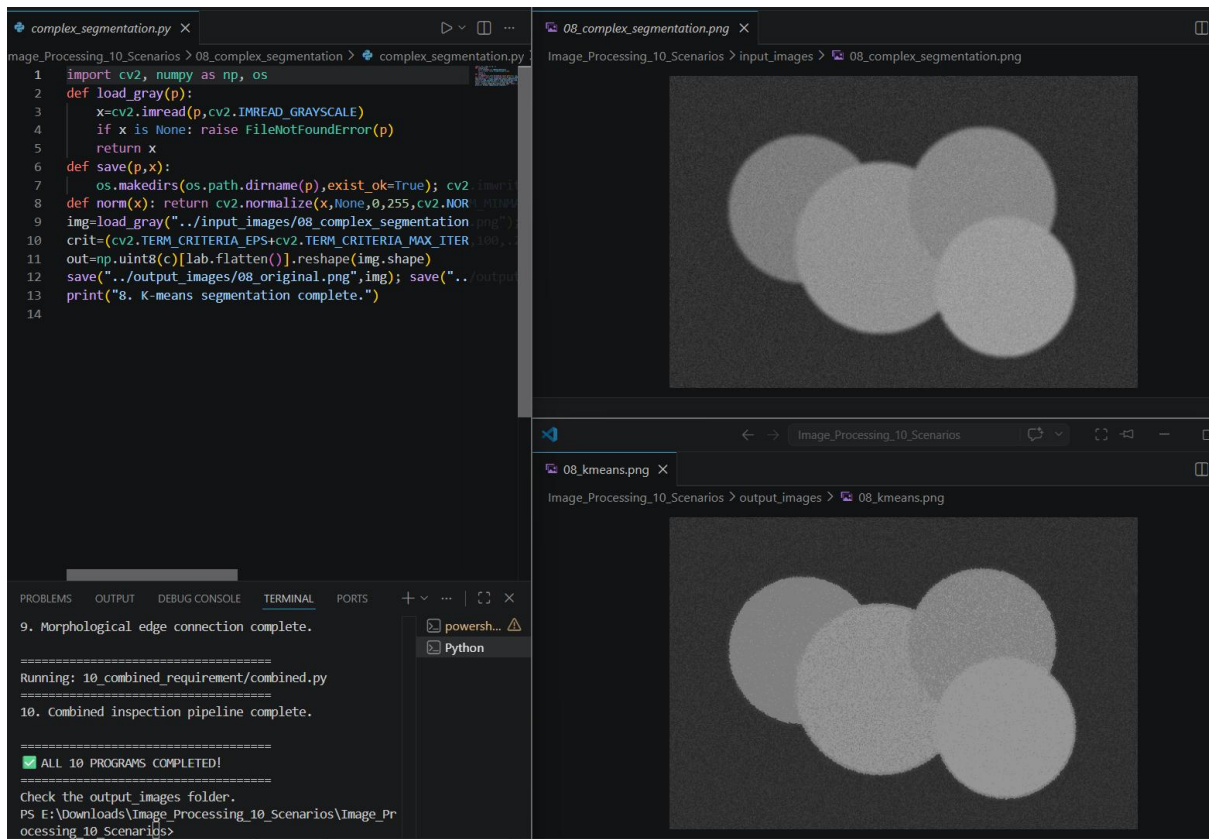
- Interpret the scenario and explain segmentation requirements.
- Identify key features enabling separation.
- Apply a suitable segmentation technique.
- Justify your decision logically.
- Present your response clearly.



8. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

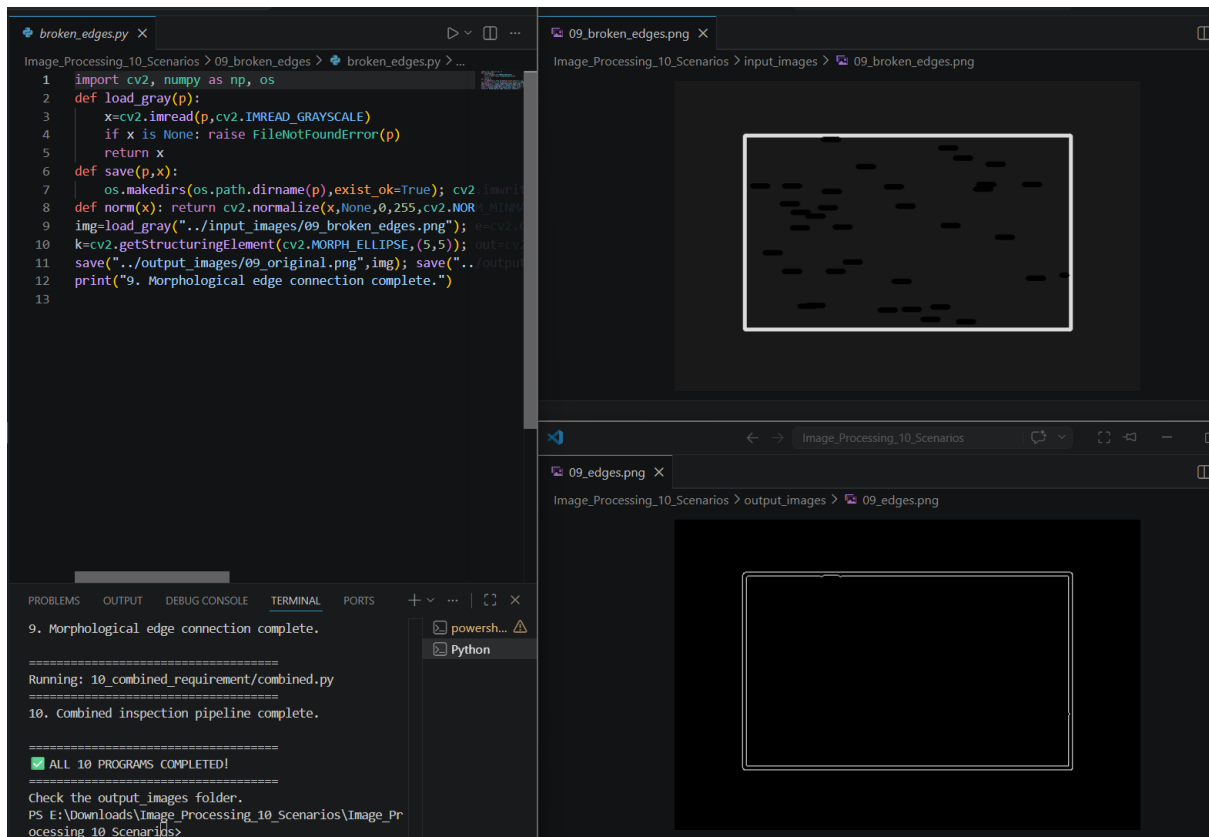
- Interpret the scenario and explain segmentation challenges.
- Identify key issues such as intensity overlap.
- Apply an appropriate advanced segmentation method.
- Justify how the method handles complexity.
- Provide a structured and clear explanation.



9. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

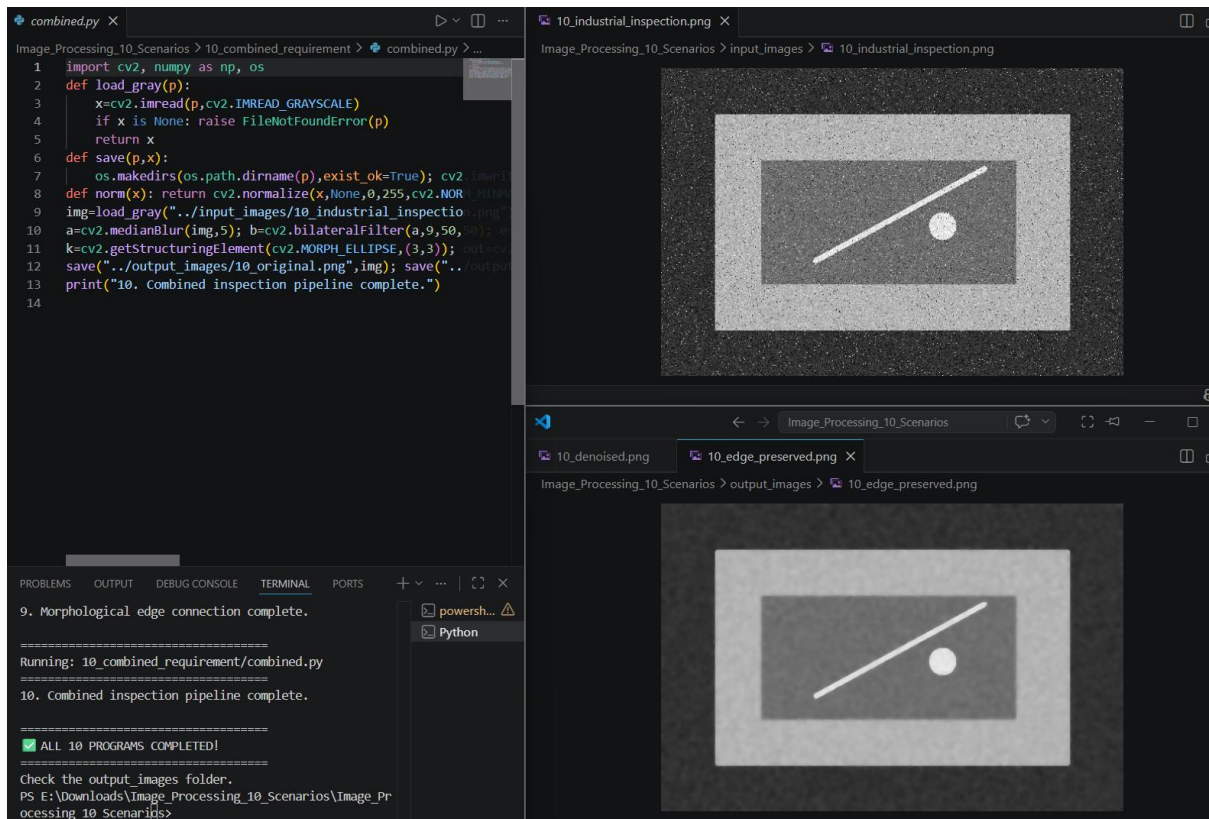
- Interpret the scenario and explain why edges are broken.
- Identify key issues affecting boundary continuity.
- Apply a suitable method to improve edge connectivity.
- Justify your approach logically.
- Present your explanation clearly.



10. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- Interpret the scenario and define the combined requirement.
- Identify key issues involving noise and edge preservation.
- Apply an appropriate sequence of techniques.
- Justify the order and selection with strong reasoning.
- Present your response in a clear and structured format.



Code of Conduct:

I S. Yuva Guru Mani Varma / 192425005 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Yuva Guru Mani Varma

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		g of the scenario and context			
Identificat ion of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Applicatio n of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well- reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Respons e	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand